

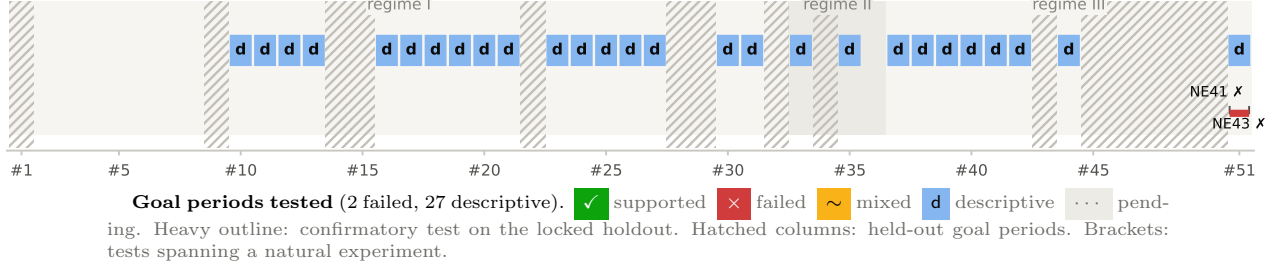
H17: Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1.5/5

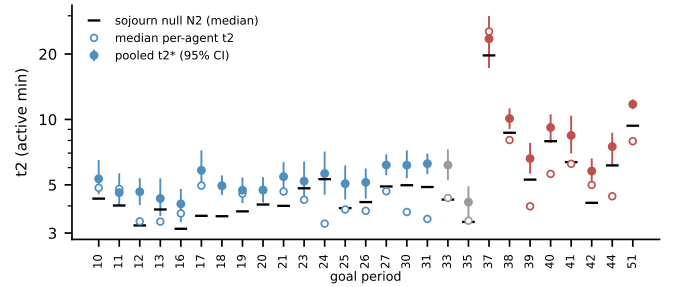
Bottom line. On action classes and Jev states, the slowest behavior mode is acting vs waiting plus agent differences, and t_2^* does not order periods by stuckness.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

Is agent behavior a Markov state model (MSM) with a few slow metastable sets? Is its slowest timescale t_2^* a per-period order parameter, longest in stuck periods? Round 1 fits MSMs on 6 action classes per active minute in 27 non-holdout periods. The Chapman–Kolmogorov (CK) test fails in 27/27. t_2^* (5–24 min) beats a sojourn null by a small margin and does not track errors. Regime III is slower than regime I. Verdicts: 13 mixed, 14 failed. Round 1b repeats the fit on Jev v3.1 behavior states (soft vectors, a noise-cancelling estimator) and counts real failures instead of stderr. The semantic slowest mode lasts hours (agent-day scale). The slow set is still acting vs waiting. Higher p_{blocked} goes with *shorter* t_2^* . The native tests (forced erasures, nudger switch-off, #27’s spontaneous shift) fail. The holdout is not run.



Observables. Slowest implied timescale per goal period at $\tau_c = 5$ min. Filled: pooled MSM (95% agent-day CI). Bars: sojourn-null median (dwell times kept, order shuffled). Open: median per-agent t_2 . The pooled MSM beats the sojourn null in every period by a small margin. The agent mixture lifts it above the typical agent. Regime III (red) is slower than regime I (blue). The #37 peak is agents idle before their first and after their last action of the day.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Nothing changes in practice for an operator: t_2^* does not detect stuck periods. The “metastable sets” are tool modality (GUI vs shell) and idle time, which any log viewer shows. Part of the GUI-vs-shell split is a difference between GUI-heavy and shell-heavy agents. The transferable result is negative and is for analysts. Minute-scale agent behavior is non-Markov, so run the CK test before you quote a timescale. A pooled MSM over different agents makes false slow modes (pooled $t_2^* \times 1.28$ the per-agent median). Soft LLM labels need a noise-cancelling estimator and $\gtrsim 15k$ windows per period. On Jev states, waiting is the only slow set.

PHYSICAL MODEL

Kinetic Potts chain out of equilibrium (models 10, 02). The state $x_t \in \{\text{browse, type, shell, chat, idle, consolidate}\}$ is one per agent-minute (H14 scheme). $C(\tau)$ counts lagged pairs within agent-days. $T(\tau)$ is the non-reversible MLE. t_k are implied timescales from eigenvalues λ_k . χ holds the PCCA+ memberships (Röblitz–Weber) of m metastable sets. $q^\pm$ are forward and backward committors, with $\delta = 0$ under detailed balance. p_t are soft Jev posteriors.

$$T_{ij}(\tau) = C_{ij}(\tau) / \sum_k C_{ik}(\tau), \quad t_k(\tau) = -\tau / \ln |\lambda_k(T(\tau))|$$

$$t_2^* \equiv t_2(\tau_c = 5 \text{ min})$$

$$\text{CK: } P_{A \rightarrow A}(k\tau) \stackrel{?}{=} [\pi_A^\top T(\tau)^k]_A$$

$$T_{\text{cg}} = (\chi^\top \Pi \chi)^{-1} \chi^\top \Pi T \chi, \quad \delta_i = q_i^+ - (1 - q_i^-)$$

$$\lambda_k^\top = \text{eig}[C(1)^{-1} C(1+\tau)], \quad C(\tau) = \sum_t p_t p_{t+\tau}^\top$$

H17-behavior-metastable-sets · generated 2026-10-04 12:03 by infra/summaries/build_summaries.py · card hypotheses/H17-behavior-metastable-sets/README.md · ratings: H17 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
P4b t_2^* tracks p_{blocked} (III)	$\rho = -0.69$; stratified -0.49 , p 0.03	(27) failed (sign)
P4b within periods	meta $\rho = -0.28$, p 0.03	(79) failed (sign)
P4b action t_2^* vs real failures	$\rho = -0.19$ (stderr: -0.43)	failed
p_{blocked} flags stuck all: periods	$\rho(\text{fail}) -0.64$; $\rho(\text{output}) -0.71$	III: mixed
P4c regime I slower	40 vs 205 min, p 0.013	reversed
P3d <i>wait</i> is its own set	5/8 regime idle+absent	III; #51: holds
P8 argmax beats R1	MSM 9/27	failed
N1 NE41 relaxation $\approx \tau \approx 21$ min vs 263 (#51)		failed
t_2^*		
N2 NE43 slower with- out drive	block-to-block swings $\times 6-7$	failed
N3 #27 change point	p 0.31	failed

GOAL PERIODS IN DETAIL

G10 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G11 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G12 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G13 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G16 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G17 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G18 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G19 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G20 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
G21 d 1b: descriptive (Jev v3.1 soft-state replication; by ...
+19 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

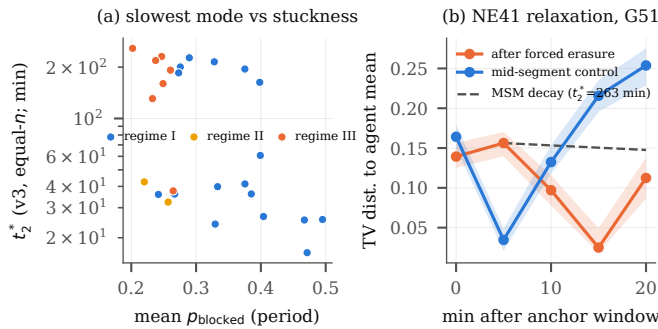
CAVEATS

Round-1b synthetic checks show a bias of $\times 1.6-2.7$ in the soft-state shifted estimator at a 4-h period's 3,600 windows. The estimator breaks down at 12 states below $\sim 15k$ windows. The soft CK test has no power, and a sojourn null on noisy argmax runs is invalid. The v3 per-period verdicts are therefore descriptive, and t_2^* comparisons use equal-size subsamples. The NE41 control anchors are selected (active, censored). p_{blocked} means “waiting on others” in regime I and “failing or looping” in regime III. Pooled MSMs mix agents (rival R2).

ROUND 2

Fit a hidden-Markov model with agent-day random effects, to separate the agent mixture from switching. Use content-level states for spirals (#45 is held out).

- **Essence.** What are the metastable cognitive modes (planning, building, debugging, existential talk), and what switches them?
- **H17-R1** A soft Jev-state MSM recovers slow cognitive modes, and switches are driven by external input.
- **H17-R2** Debugging is the slowest mode and the main sink of time.
- **H17-R3** Mode switches synchronize across agents only through shared artifacts (E3).



Second figure. (a) Semantic t_2^* (equal sample size) vs the period's mean p_{blocked} . It is bimodal (tens of minutes or a day length) and tends to be shorter where agents are more blocked. (b) After forced erasures (#51), the state deviation decays in ~ 20 min. The MSM's t_2^* predicts the dashed line. The control rises because of censoring selection.